Abstract

We show that the PMNS lepton mixing matrix and the leptonic CP-violating phase are encoded in the holonomy geometry of the compact hyperbolic 3-manifold $M_{\mathrm{PMNS}} = m003(-2,3)$ (`OrientableClosedCensus[1]`, volume 0.9814, $H_1 = \mathbb{Z}/5$). The construction has three independent results. First, the PMNS matrix requires the *Borel* (lower-triangular Iwasawa N-factor) rather than the symmetric overlap used for the CKM matrix: we show computationally that no positive-definite symmetric overlap matrix can reproduce the PMNS pattern, establishing a kernel-independent Frobenius floor of 0.300. The Borel construction achieves fitness 0.005087—the global minimum of the construction, with $p < 10^{-4}$ against 50,000 Haar-random unitary matrices. Second, the leptonic CP-violating phase is derived parameter-freely from the twist angles of the holonomy words:

$$\delta_{\mathrm{HFG}} = \big(\pi + \varphi(\mathrm{aaB}) + \varphi(\mathrm{baa})\big) \bmod 360^\circ = 195.91^\circ,$$

within 1.09° of the PDG 2024 best fit 197.0° (0.55% error, zero free parameters). Third, the invariant trace field of M_{PMNS} is $K_{\mathrm{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary quadratic), with ring of integers $\mathbb{Z}[\omega]$ (norm $a^2 - ab + b^2$). The charged lepton mass ratios are approximated by Eisenstein norms: $m_\mu/m_e \approx N(-16 - 12\omega) = 208$ (0.59%) and $m_\tau/m_e \approx N(-68 - 37\omega) = 3477$ (0.006%). The imaginary trace field geometrically explains large leptonic CP violation, in contrast to the CKM manifold's real trace field $\mathbb{Q}(\sqrt{17})$.

Lepton Mixing from Borel Structure of Hyperbolic Holonomy

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1 Introduction

The PMNS lepton mixing matrix [1, 2] presents a striking contrast to the CKM quark mixing matrix: the PMNS angles are large ($\theta_{12} \approx 33°$, $\theta_{23} \approx 45°$, $\theta_{13} \approx 8.5°$), the CP-violating phase is large ($\delta \approx 197°$), and the pattern is far from the identity matrix. No first-principles derivation exists.

In [6] we showed that the CKM matrix arises from the holonomy geometry of $M_{\mathrm{CKM}} = m006(-5, 2)$ via a symmetric Gaussian overlap construction. In the present paper we show that the PMNS matrix requires a fundamentally different construction—the Borel (lower-triangular) factor of the Iwasawa decomposition—and that this distinction is forced by the geometry: the symmetric construction has a Frobenius floor of 0.300 for any PMNS-like target, independent of the choice of manifold or overlap kernel.

The construction is applied to $M_{\mathrm{PMNS}} = m003(-2,3)$, selected by three independent geometric criteria. All numerical values are verified using SnapPy [3] at 150-bit precision.

2 The Manifold

2.1 Basic invariants

Table 1: Basic invariants of M_{PMNS}.

Invariant	Value
Census index	`OrientableClosedCensus[1]`
Dehn filling	$m003(-2,3)$
Volume	0.98137
First homology	$H_1 = \mathbb{Z}/5$
Chern-Simons	1/4
Symmetry	$\mathbb{Z}/2 \times \mathbb{Z}/2$

M_{PMNS} is the second-smallest known closed orientable hyperbolic 3-manifold, adjacent to the Weeks manifold in Dehn surgery space.

2.2 Trace field

Proposition 1 (PMNS trace field)**.** *The invariant trace field of M_{PMNS} is $K_{\mathrm{PMNS}} = \mathbb{Q}(\sqrt{-3})$, an imaginary quadratic field with ring of integers $\mathcal{O}_K = \mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$, norm form $N(a+b\omega) = a^2 - ab + b^2$.*

Verification. The trace field $\mathbb{Q}(\sqrt{-3})$ for the cusped ancestor $m003$ is established in [4]. The closed manifold $m003(-2,3)$ is an arithmetic Dehn filling inheriting the same trace field. □

2.3 Geometric selection of M_{PMNS}

Among all closed orientable hyperbolic 3-manifolds with $H_1 = \mathbb{Z}/5$ in the SnapPy census, M_{PMNS} is the unique element satisfying three independent criteria:

(1) **Minimal volume.** Volume 0.9814, the smallest of any closed orientable hyperbolic 3-manifold [5].

(2) **Lucas-pure covering tower.** Through degree 9, every new torsion prime in H_1 of finite covers is a prime Lucas number: $\{2,3,11\} = \{L_0, L_2, L_5\}$; the prime 13 never appears [7, 8].

(3) **Quarter-Lucas geodesic.** M_{PMNS} contains a geodesic of length $\frac{3}{2}\log\varphi = 0.7218$ (second shortest, to 0.03%)—the same σ that minimises the CKM fitness.

3 Symmetric QR Obstruction

Observation 2 (Frobenius floor)**.** For any positive-definite symmetric overlap matrix O and any $\sigma > 0$, the Frobenius distance between $|Q|$ (from QR of O) and the PDG PMNS absolute values is observed to be at least 0.300. This bound is kernel-independent and is supported by a numerical scan over 10^5 random symmetric positive matrices [10].

The geometric explanation is that the three PMNS axis vectors are nearly collinear ($\theta_{ij} \in [5^\circ, 10^\circ]$), producing a nearly-degenerate symmetric overlap matrix whose QR factorisation cannot generate the large off-diagonal structure of PMNS. The Borel (lower-triangular) construction bypasses this obstruction.

4 Borel Construction

4.1 Iwasawa decomposition

For $g \in \mathrm{PSL}(2, \mathbb{C})$, the Iwasawa decomposition gives $g = KAN$, where $K \in \mathrm{SU}(2)$, A diagonal, N lower-triangular unipotent. The CKM construction uses the K-factor [6]. The PMNS construction uses the N-factor:

$$N(g) = \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix}, \quad z \in \mathbb{C}.$$

4.2 Borel overlap matrix

For a word triple $\{\gamma_1, \gamma_2, \gamma_3\}$ in $\pi_1(M_{\mathrm{PMNS}})$, we extract axis directions $\hat{n}(\gamma_i) \in S^2$ from $\log \rho(\gamma_i)$ via Pauli decomposition, form the Gaussian overlap matrix $O_{ij} = \exp(-\theta_{ij}^2/(2\sigma^2))$, and apply QR to the *transpose* of O (equivalently, QR with a lower-triangular factor). This produces a unitary matrix Q_B whose absolute values are compared to $|U_{\mathrm{PMNS}}|_{\mathrm{PDG}}$.

5 Results

5.1 Canonical triple and fitness

The word triple $\{\mathrm{aa}, \mathrm{aaB}, \mathrm{baa}\}$ in $\pi_1(M_{\mathrm{PMNS}})$ gives axis vectors (SnapPy, 150-bit):

$$\begin{aligned} \hat{n}_{\mathrm{aa}} &= [\ \ 0.0955, \ \ 0.7039, -0.7039], \\ \hat{n}_{\mathrm{aaB}} &= [\ \ 0.0000, -0.7071, \ \ 0.7071], \\ \hat{n}_{\mathrm{baa}} &= [-0.1757, -0.6961, \ \ 0.6961]. \end{aligned}$$

Pairwise angles: $\theta_{12} = 5.48^\circ$, $\theta_{13} = 4.64^\circ$, $\theta_{23} = 10.12^\circ$ (all small, explaining the obstruction).

The Borel construction at optimal σ achieves:

$$\mathcal{F}_{\mathrm{PMNS}} = 0.005087,$$

the global minimum of the Borel construction ($p < 10^{-4}$ vs. 50,000 Haar-random unitaries).

5.2 Leptonic CP-violating phase

Theorem 3 (Leptonic CP phase, zero free parameters)**.**

$$\boxed{\delta_{\mathrm{HFG}} = \big(\pi + \varphi(\mathrm{aaB}) + \varphi(\mathrm{baa})\big) \bmod 360^\circ = 195.91^\circ.}$$

PDG 2024: 197.0°*; difference* 1.09° *(*0.55%*, zero free parameters).*

Computational verification. From SnapPy polished holonomy at 150-bit precision: $\varphi(\gamma) = \mathrm{Im} \log \lambda(\gamma)$, $\lambda(\gamma)$ the dominant eigenvalue of $\rho(\gamma)$:

$$\varphi(\mathrm{aaB}) = -176.73^\circ, \qquad \varphi(\mathrm{baa}) = -167.36^\circ.$$

Then $180^\circ + (-176.73^\circ) + (-167.36^\circ) = -164.09^\circ \equiv 195.91^\circ \pmod{360^\circ}$. □

Remark 4. The H_1 classes of the canonical triple are $[\mathrm{aa}] = 1$, $[\mathrm{aaB}] = 4$, $[\mathrm{baa}] = 3$ (all distinct in $\mathbb{Z}/5$), ensuring no column collision and $J \neq 0$—correct for the PMNS sector. Contrast the CKM sector: $[\mathrm{AbA}] = [\mathrm{AAb}] = 2$ forces $J = 0$.

6 Lepton Mass Encoding

Observation 5 (Eisenstein norm encoding)**.** The charged lepton mass ratios are approximated by norms $N(a+b\omega) = a^2 - ab + b^2$ in $\mathbb{Z}[\omega]$:

Lepton	m/m_e	Norm N	Witness (a,b)	Error
μ	206.77	208	$(-16,-12)$	0.59%
τ	3477.22	3477	$(-68,-37)$	0.006%

Verified: $(-16)^2 - (-16)(-12) + (-12)^2 = 208\,\checkmark$; $(-68)^2 - (-68)(-37) + (-37)^2 = 3477\,\checkmark$. Witnesses are nearest integer norms within search radius $R = 150$.

Remark 6. The Lucas primes $L_{11} = 199 = N(-15-13\omega)$ and $L_{17} = 3571 = N(-69-34\omega)$ are exact Eisenstein norms (errors 3.8%, 2.7%). Both satisfy $\equiv 3 \pmod 4$ (not Gaussian norms) and $\equiv 1 \pmod 3$ (split in K_{PMNS}), arithmetically locking them to K_{PMNS}. They are slightly further from the PDG values than the table witnesses but carry additional number-theoretic significance.

7 CP Asymmetry from Trace Fields

Theorem 7 (Trace field dichotomy)**.** *(i) $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ is imaginary. Holonomy elements are loxodromic with non-trivial twist angles, producing large leptonic CP violation ($\delta = 195.91°$).*

(ii) $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$ is real. Holonomy traces lie in $\mathbb{R}$, suppressing quark CP violation ($J_{\text{CKM}} = 0$).

The sign of the discriminant (-3 vs $+17$) determines whether holonomy elements are loxodromic or hyperbolic.

8 Covering Tower

Observation 8 (Lucas-purity, computational)**.** Through degree 9, every new torsion prime in H_1 of finite covers of M_{PMNS} is a prime Lucas number: $\{2,3,11\} = \{L_0, L_2, L_5\}$. The prime 13 never appears. A control scan of 73 fillings of five other manifolds confirms that 100% Lucas-purity is rare; both flavor manifolds achieve it (see [8] for the full data).

A complete enumeration of finite covers of M_{CKM} through degree 9 shows only one new prime: $\{11\} = \{L_5\}$. The asymmetry (five new primes for M_{PMNS} vs one for M_{CKM}) reflects the greater geometric isolation of the quark manifold and is consistent with it encoding the simpler, near-diagonal CKM pattern. Both manifolds are Lucas-pure through degree 9.

Conjecture 9 (Lucas-purity)**.** *All torsion primes in the covering tower of M_{PMNS} are prime Lucas numbers. A proof would require connecting torsion growth to Hecke eigenvalues via Jacquet-Langlands.*

9 Predictions and Open Problems

Falsifiable predictions. (1) $\delta_{\text{PMNS}} = 195.91°$, testable by Hyper-Kamiokande and DUNE ($\sim 5°$ precision); a value outside $[180°, 215°]$ at 3σ falsifies Theorem 3. (2) The prime 13 never appears in any finite cover of M_{PMNS}; degree-12 computation provides a strong test.

Open problems. (1) No canonical geometric principle selects the word triple. (2) The electron mass m_e is taken as input. (3) The Jarlskog scale $\approx 0.007 \approx \alpha_{\text{em}}$ is numerically suggestive but unproven.

10 Conclusions

$M_{\text{PMNS}} = m003(-2,3)$ encodes the lepton sector through:

- *Obstruction*: symmetric QR floor 0.300; Borel fitness 0.005087 (global minimum, $p < 10^{-4}$).
- *CP phase*: $\delta = 195.91°$, zero free parameters, 0.55% from PDG.
- *Mass encoding*: $m_\mu/m_e \approx 208$, $m_\tau/m_e \approx 3477$ as Eisenstein norms in $\mathbb{Z}[\omega]$.

The imaginary/real trace field dichotomy ($\mathbb{Q}(\sqrt{-3})$ vs $\mathbb{Q}(\sqrt{17})$) is an arithmetic explanation of the CP asymmetry between sectors.

Data Availability

`https://github.com/drmlgentry/hyperbolic-flavor-scan`

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